

Isabella Banse

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EDUCATION

University of Colorado Boulder

Bachelor of Science in Computer Science

Boulder, CO

Aug. 2024 – Present

- GPA: 3.909, *Dean's List*
- CU Boulder Chancellor's Achievement Scholarship

EXPERIENCE

University Educator

Jan. 2025 – May 2025

Partnerships for Informal Science Education in the Community

Boulder, CO

- Conducted weekly STEM outreach at a middle school, teaching thermodynamics through hands-on experiments to engage underrepresented youth
- Adapted lessons to different learning styles, using clear communication and patience to support students in developing their STEM identity
- Collaborated with educators and peers to refine curriculum effectiveness through meetings and detailed field notes

Cashier

Oct. 2025 – Jan. 2026

Sewall Market

Boulder, CO

- Processed steady-flow transactions accurately and efficiently while delivering customer service in a fast-paced campus retail environment
- Tracked and documented inventory levels, identified stock shortages, and supported restocking processes to improve product availability and workflow efficiency
- Managed multiple digital payment systems (credit/debit, campus cash, and meal plan platforms), ensuring accurate transaction processing and system consistency

PROJECTS

EEG Genre Classifier Neural Network | *Python, TensorFlow, Keras, NumPy, GitHub*

Mar–Apr. 2026

- Built and trained a feedforward neural network using TensorFlow/Keras to classify music genres from simulated EEG brain wave data, achieving 93.8% accuracy on held-out test data
- Engineered a synthetic EEG dataset of 1,600 samples across 4 genres and 2 participant groups using NumPy, modeling genre-specific alpha and beta wave distributions based on neuroscience research
- Trained and evaluated separate models for musician and non-musician groups, identifying a 23-point accuracy gap and conducting cross-group generalization testing to analyze neurological differences between populations

FitX Fitness Tracker | *HTML, SQL, JavaScript, Handlebars, CSS, Docker, GitHub, Bootstrap*

Nov. 2025

- Developed a full-stack fitness tracking web application enabling users to log workouts, manage friends, and monitor nutrition data through an SQL database system
- Engineered backend integration of an external nutrition API to parse and store data for dynamic user retrieval
- Collaborated in a 5-member team using Git and GitHub for version control, branch management, and pull request reviews to coordinate feature development and maintain code quality

RELEVANT COURSEWORK

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|------------------------|------------------------------|-----------------------------------|
| • Algorithms | • Linear Algebra | • Physics I |
| • Software Development | • Discrete Structures | • Calculus II |
| • Computer Systems | • Human-Computer Interaction | • Calculus I |
| • Data Structures | • Intro to CS Research | • CS as a Field of Work and Study |

TECHNICAL SKILLS

Languages: C/C++, Java, Python, JavaScript, HTML, Handlebars, SQL, LaTeX, CSS

Developer Tools: Git, Docker, Visual Studio Code, Postman, Node.js

EXTRACURRICULARS

- Chi Omega Fraternity, Society of Women Engineers, CU Women in Computing, CU Best Buddies, University United Methodist Church Volunteer